

ASCENT instructions

Absolute Single-cell Copynumber Evaluation and subclonE Topology

ASCENT is an end-to-end method for direct-tagmentation base single-cell WGS (and scRNA-seq).

ASCENT is developed to identify subclonal CNVs, call absolute and allele specific copy numbers on data derived from **direct-tagmentation** based protocols.

Set up

1. Clone the repository to your local machine:

```
git clone https://github.com/Engelab/ASCENT.git
```

2. Create the ASCENT environment

```
conda env create -f workflow/envs/ASCENT.yaml
```

3. Download necessary genome files

Grch38:

<https://gdc.cancer.gov/about-data/gdc-data-processing/gdc-reference-files>

Blacklist:

<https://www.encodeproject.org/files/ENCFF356LFX/>

If RNA will be processed a star index should be built

4. Update the following paths in `configs/myconfigfile.yaml` with the location of the genome files

fasta: /wrk/resources/genomes/grch38-gdc/grch38.d1.vd1.ercc.fa

genes: /wrk/resources/genomes/hg38-iGenome-ERCC/hg38_RefSeq.bed

rna_ref_dir: /wrk/resources/genomes/grch38-gdc/star_index

rna_gtf: /wrk/resources/genomes/grch38-gdc/gencode.v36.filtered.gtf

chr_list_intersect: /wrk/resources/genomes/grch38-gdc/grch38.d1.vd1.ercc.genome

genome_bed: /wrk/resources/genomes/grch38-gdc/grch38.d1.vd1.ercc.bed

blacklist: /wrk/resources/genomes/grch38-gdc/grch38_encode_exclusion_list.bed

maptrack: /wrk/resources/genomes/grch38-gdc/grch38.k36.umap.sorted.bed

gene_info: /wrk/resources/genomes/grch38-gdc/star_index/geneInfo.tab

mt_genes: /wrk/resources/genomes/grch38-gdc/gencode.v36.mt_genes.txt

Starfusion_ref:

/wrk/resources/genomes/GRCh38_gencode_v44_CTAT_STARFusion/ctat_genome_lib_build_dir

Make a seedfile

A seedfile lists the cell_id in the first column, fastq read 1 in the second and fastq read 2 in the third. Column names should be: cell, fq1, fq2.

Adjust the config file

See config/HCT.yaml as an example.

Path to the DNA seedfile and an out directory are required.

Under **paths** you should list in which folder fastq files are found (**dnain**) and where bam files, count files and other single cell files should end up (**dnaout**).

The paths for reference files should have been updated during **Set up**

Under **DNA** there are multiple parameters that can be changed. The most important ones are:

- Binsize: For a normal ultra-low coverage data we run 10-40kb binsize (this is the binsize that original clustering and single cell segmentation happens at).
- Binsize_refine: should most often be 10.000. This is the binsize that clone refinement will occur on.
- Normals_bins: If no normal cells have been sequenced with the same method and same sequencer then normals_bins should be left empty. If normal cells have been sequenced then this should be a seedfile indicating where the fastq files are. These cells will be used to further filter the bins used for analysis, based on method/sequencer specific artefacts.
- Normals_scaling: If no normal cells have been sequenced with the same method and same sequencer this should be left empty. If normal cells have been sequenced then this should be a seedfile indicating where the fastq files are. These cells will be used to normalize count matrices for mpcf and plotting if normalize_to_panel is set to TRUE
- Normalize_to_panel: This indicates if count matrices should be normalized to normal cells sequenced on the same sequencer with the same method.
- Gamma: This is the penalty that mpcf on single cell level is run, before initial clustering is performed. It is best to test 3-5 gammas per case, most often 1,2,3,4 and 5 are good for most cases.

Run snakemake

To run ASCENT use

```
snakemake --configfile=config/MyConfig.yaml --use-conda
```

This will run all steps of the pipeline. A good tip is to start with a dry run by adding -n to the command to see which processes will run. Limit cores according to your setup. We commonly run --cores 12.

The default behaviour of the pipeline is to run mpcf segmentation and output clones, refined segments, absolute allele specific copy numbers and a medicc2 tree. If the hypothesis free segmentation, based on llr, is to be run that rule should be targeted specifically, either by adding `--until concat_copynumber_llr` to the snakemake command or by targeting a output file from those scripts, found in rule `dna_llr.smk`.

Analyse result

Cell specific results will be found in folder `dnaout/cell_id/`

```
SKE00003D_A03_bed.gz      SKE00003D_A03-bincounts-00000.tsv      SKE00003D_A03_dedup.bam      SKE00003D_A03-picard-inserts.pdf      SKE00003D_A03-picard-mark_duplicates.txt      trimmed/
SKE00003D_A03-bincounts-10000.tsv      SKE00003D_A03-chr_counts.txt      SKE00003D_A03_dedup.bam.bai      SKE00003D_A03-picard-inserts.txt      SKE00003D_A03.scp.bed.gz
```

Other results will be in the output directory from the config file

There are three output folders, **QC** where all QC is gathered, **clones** where plots from clonal analysis can be found and **medicc**, where the medicc output is found.

Are my clones accurate?

The most important output files to look at are the heatmaps `results/*/clones/*final-refined-clones-heatmap-g*-b*-br*.png` and the pdf which shows a clonal overview over each chromosome, labeling all segments, in `results/*/clones/*final-clones-refined-g*-b*-br*.pdf`

Based on the pdf it is easy to see if breakpoints are correctly identified or if the segmentation should have had a higher or lower penalty (gamma).

!!When working with complex samples it can happen that the default parameters do not give satisfactory clones or segments

If this happens interactive clone refinement is possible with the script:

```
workflow/rules/scripts/snippets/ManualCloneRefinement.R
```

KEEP IN MIND that if the data that goes into clone refinement has been clustered suboptimally it is often easier to alter the assignments at that step than during refinement. We recommend running different gammas until the heatmap from the rule `"find_clones"`

```
results/*/clones/*-heatmap_revised-umap-g*-*-.png
```

looks satisfactory. This is easily achieved by running the snakemake command with `--until find_clones` with multiple different gammas in the config file.

It can also be worth looking at the scripts in `workflow/rules/scripts/snippets/*` to estimate the correct gamma and noise level.